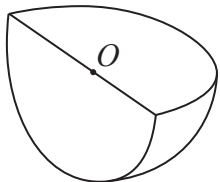


19 Shape **S** is one quarter of a solid sphere, centre O .



Shape **S**

The volume of **S** is $576\pi \text{ cm}^3$

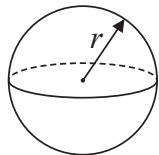
Find the surface area of **S**.

Give your answer correct to 3 significant figures.

You must show your working.

$$\text{Volume of sphere} = \frac{4}{3}\pi r^3$$

$$\text{Surface area of sphere} = 4\pi r^2$$



cm^2

(Total for Question 19 is 5 marks)